

STSM3764 – Generalised Linear Models

Class Times

- Monday 11:10-12:00 S1
- Tuesday 14:10-16:00 WWG114
- Wednesday 12:10-13:00 FG/PH116

Method and Assessment

- Portfolios and interviews (mid-semester and end-of-semester assessments) [30%, then 50%]. **Read the portfolio creation document**
- 1 practical assignment encompassing all module content, submitted in stages over the semester and incorporated into the P/F [20%]
- Weekly class tests [10%]

Learning Outcomes for Generalised Linear Models

1. Generalising the linear model
2. Estimation
3. Inference
4. Binary data and logistic regression
5. Poisson regression and log-linear models
6. Data with constant coefficient of variation

Student will be able to:

- demonstrate knowledge of the theory of generalised linear models,
- formulate models with the appropriate choice of error distribution, link function and variance function for particular data sets,
- reflect on the merits of alternative model formulations for relevant data sets,
- compare generalised linear models, linear models and contingency table methods,
- demonstrate knowledge of ML estimation for generalised linear models,
- perform hypothesis testing and interval estimation in generalised models,
- apply and analyse goodness of fit statistics for generalised linear models,
- evaluate relevant data sets by fitting generalised linear models using statistical software, and interpret the results.

Actuarial outcomes (we probably won't have time to cover all of these)

2.1.4 Use Principal Components Analysis to reduce the dimensionality of a complex data set

4.2 Generalised linear models

4.2.1 Define an exponential family of distributions. Show that the following distributions may be written in this form: binomial, Poisson, exponential, gamma, normal.

4.2.2 State the mean and variance for an exponential family, and define the variance function and the scale parameter. Derive these quantities for the distributions above.

4.2.3 Explain what is meant by the link function and the canonical link function, referring to the distributions above.

4.3.4 Explain what is meant by a variable, a factor taking categorical values and an interaction term. Define the linear predictor, illustrating its form for simple models, including polynomial models and models involving factors.

4.2.5 Define the deviance and scaled deviance and state how the parameters of a generalised linear model may be estimated. Describe how a suitable model may be chosen by using an analysis of deviance and by examining the significance of the parameters.

4.2.6 Define the Pearson and deviance residuals and describe how they may be used.

4.2.7 Apply statistical tests to determine the acceptability of a fitted model: Pearson's chi-square test and the likelihood ratio test

4.2.8 Fit a generalised linear model to a data set and interpret the output.

What I really want to do:

- Talk about GLMs What, why, how, when?
- Look at real data, and test various GLMs.
- Interpret the model outputs

Student will be able to:

- **demonstrate knowledge of the theory of generalised linear models,**
 - Can you write a distribution in exponential form, identifying and naming the exponential family components?
 - What is the implication of a link function being "canonical"?
 - Can you estimate the mean and the variance of the sufficient statistic from this form? **[2 weeks]**
- **formulate models with the appropriate choice of error distribution, link function and variance function for particular data sets,**
 - Can you build a model to explain/predict a continuous/binary/[*categorical/ordinal/constant CV*] variable, with factors and continuous variables, including interactions and possible polynomial terms, using the appropriate canonical link function?
 - [*Can you discuss the use of canonical vs non-canonical links in GLMs.*]
- **reflect on the merits of alternative model formulations for relevant data sets,**
 - What would a Normal OLS interpretation be for a naïve model?
 - How does our interpretation differ as we choose a more correct model?
- **compare generalised linear models, linear models and contingency table methods,**
 - Can you compare P/B/N/G models, [*as well as log-linear models, ordinal models, multinomial models and models with tolerance distributions*]?
 - Do you know how to interpret the coefficients for the different models?
- **demonstrate knowledge of ML estimation for generalised linear models,**
 - For P/B/N/G can you derive the MLE of the distribution parameter?
 - For P/B/N/G can you derive the score equations for the regression parameters, and the Fisher Information matrix, and describe the process for estimating the parameter MLEs?
- **perform hypothesis testing and interval estimation in generalised models,**
 - For P/B/N/G can you derive the deviance?

- Do you know what a LR test is, and what it can test? How do you perform an LR test with deviances?
 - Do you know what a Wald test is, and what it can test? How do you test a contrast with a Wald test? How do you perform a Wald test? How does the Wald test compare with the LR test?
 - How do you estimate parameter intervals using the LR or Wald test?
- **apply and analyse goodness of fit statistics for generalised linear models,**
 - What are the null and residual deviances? How are they used?
 - *[What types of residuals are there for GLMs, and how would we analyse them in terms of GoF?]*
 - What other GoF statistics are there, and how do you use each one of them?
 - How do you analyse a GLM by assessing predictive fit?
- **evaluate relevant data sets by fitting generalised linear models using statistical software, and interpret the results.**

Additional interview questions

- Why would we choose to use a GLM instead of a classical OLS linear regression?
- Why is it important for us to determine whether or not a distribution is from the exponential family?
- How do you determine whether or not a distribution belongs to the exponential family?
- What are the main components of a GLM?
- How do you find the link function for a GLM?
- What does it mean when the link function is canonical?
- Is there any reason to use non-canonical link functions?
- Why use canonical?
- What is the score statistic for a GLM's slope parameter?
- How does one determine the GLM MLE for the linear predictor's parameters? Can you describe two ways?
- How does one interpret the slope parameter estimates for:
 - An identity link
 - A log link
 - A logit link
 - An inverse link
- How can we **test** goodness of fit for a single GLM?
- How can we **test** multiple parameters simultaneously in a GLM?
- What are the similarities and differences between an Ordinal Logit and Multinomial Logit model?
- What does the dispersion parameter in a GLM signify? How is it related to the variance?

Cheating

- If you try to hand in someone else's portfolio, I will send you to a disciplinary hearing AND give you 0 for the module.
- You are allowed to work in groups, IF you mention who you worked with, on what sections you worked together, and what you contributed.

Key terms

- Generalised linear model
- Exponential family
- Link function
- Mean or inverse-link function
- η – eta
- Linear predictor
- Parameters
- Natural parameter
- Sufficiency
- Sufficient statistics
- Dispersion parameter
- Log-partition function
- Score function
- Maximum likelihood estimates
- Canonical form
- Probit
- Logit
- Nuisance parameter

Distributions for data:

- Normal
- Bernoulli
- Binomial
- Multinomial
- Ordinal
- Negative Binomial
- Poisson
- Exponential
- Gamma

2022 Some structure for the ‘normal’ students

Week 1

- Read about 2 chapters of the 2012 Prof Schall’s notes
- Read Wikipedia “Generalized Linear Models”
- Read 2003 Jordan chapter
- Start writing stuff down
- Aim for LOs up to MLE (and 1st draft of practical assignment) to be in your P/F by mid-September, for 1st portfolio interview session, starting 13 September.

Week 2

- Start looking at 1-parameter exponential family. Do the math! Factorise your known distributions into exponential form $f(x|\theta, \tau) = h(x, \tau) \exp \left[\frac{\eta^T(x) - A(\eta)}{d(\tau)} \right]$

- Some notation: τ or ϕ for the dispersion parameter, ξ for the linear predictor. If $T(x)$ is the identity function, $T(x) = x$, then your exponential form is **canonical**, and η is the “natural” parameter

We are relating a dependent variable (of some sort) to a set of independent variables.

- Y has some distribution – this distribution is characterized by ONE PARAMETER, and is one of the exponential family of distributions. $E[Y|X] = \mu$
- $X\beta$ is some sort of linear predictor $\xi = X\beta$
- And we have some sort of link between the two. $f(\xi) = \mu$
- Our distribution has a parameter, and a *natural* parameter: η .
- Gaussian/Normal OLS is a special form of GLMs. $Y = X\beta + \epsilon, \epsilon_i \sim N(0, \sigma^2), i. i. d.$
 - This is the same as saying: $\mu = X\beta$, Y has a Normal distribution $Y \sim N(\mu, \Sigma)$ is simply diagonal with elements ϵ_i . Our link between the linear predictor and the mean of the response variable is the identity function.

Ethics

Using prohibited material in assessments, copying another student’s work, or allowing someone to copy your work on any evaluation in any way will lead to a mark of 0 for the entire assessment and might result in further disciplinary action or even an incomplete or fail for this module. Offenders might be placed on a list of No Confidence. These candidates will not receive the endorsement of this department when attempting to register as a member of ASSA, i.e. we will not sign the recommendation forms needed to register. Please always do your own work.

In completing any assignment, students are expected to produce their work independently without using artificial intelligence (AI) tools. Any use of AI tools, including but not limited to AI-generated content, is prohibited. Violation of this instruction may lead to academic misconduct.